

COURSE CODE/TITLE: CPE 031-CPE21S1 - Visualizations and Data Analysis

SECTION: CPE21S1

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NAME: Mendoza, Roy Gabriel I.

INSTRUCTOR: Engr. Jimlord Quejado

ACTIVITY NO. 2 Hands-on Activity 2 | Interactive Widget

1. PROCEDURE OUTPUT

Part 1:

The screenshot displays a web browser interface with three main sections. On the left, the 'User Profile' for Roy Gabriel Mendoza is shown, including his name, student status at the Technological Institute of the Philippines, and contact information. The middle section, titled 'Course Outline', lists four modules: 'Module 1: Experience Analytics' (100% complete), 'Module 2: Data Collection and Storage' (83% complete), 'Module 3: Artificial Intelligence and Machine Learning' (0% complete), and 'Module 4: Embarking on Your Career in Data Analytics' (100% complete). The right section, 'Data Pipeline Game', features a 'Select an option' prompt and three buttons: 'Read JSON', 'Read XML', and 'Read CSV'. A data pipeline diagram is visible in the background of the game section.

Reflection:

On this part, I was able to completely identify what is the differences from all the data formats from the activity. I have also learned the meaning of each name like the **CSV**, which means that it is a **Comma Separated Values**, or like the one in the example picture which is the **XML**, or the **Extensible Markup Language**, which has a structure containing each word, phrases or contents. These formats have different use like the one where they came from from the activity, for example the **XML** format came from a **Vending Machine**, meaning each or different machines uses different formats that they might easily be used for that particular machine.

Part 2:

The screenshot displays a web application with two main sections. On the left is a user profile for ROY GABRIEL MENDOZA, a BSCE student at the Technological Institute of the Philippines. The right section shows a 'Cisco Networking Academy' course titled 'CPE 031-CPE21S1 - 1 - 2627'. The course outline lists four modules: Experience Analytics (100%), Data Collection and Storage (83%), Artificial Intelligence and Machine Learning (0%), and Embarking on Your Career in Data Analytics. The 'Data Collection and Storage' module is expanded, showing sub-topics like '2.2.1 Data Pipelines' and '2.2.2 Lab - Interactive Widget Lab 2'. The lab interface features a data table with columns 'n', 'x', 'y', and 'z', and a 'Read JSON' button. The table contains 7 rows of data, with some values appearing to be corrupted or missing. To the right of the table are three buttons: 'Fix Corrupted Rows', 'Remove Duplicates', and 'Delete Empty Rows'.

n	x	y	z
1	-0.9	-0.5	1.1
2	-1.5	-0.2	1.5
3	-1.2	1	0.9
4			
5	-1.1	0.3	-1.1
6	-1.8	1.8	-1.3
7	0.9	-0.5	-0.5

Reflection:

In this part, we are able to identify what kind of “**corruption**” or data error is the problem in each file. From what I know, each data decryption or data conversion from the format form to the readable form, there is a slight chance that there would be an error or a problem that might arise like **Duplication**, **Corruption**, or just **Empty data information**, may it be an **empty list**, **row**, or **column**. These errors must be fixed very immediately so that there would not be next information that might cause also an error, I do not mean that it is contagious, but the computer language is procedural, what I mean is that since the first part is affected, there is a slight chance that the next part would also be affected due to the previous problem.

Overall, in this part, I have learned that we must fix the errors we found in the decryption of the format forms

to readable forms, since it could hinder our understanding of the full scope of the information from the data that we have.

Part 3:

The screenshot displays a web application with a user profile on the left and a data pipeline game on the right. The user profile for ROY GABRIEL MENDOZA includes a profile picture, name, and contact information. The data pipeline game, titled '2.2.2 Lab - Interactive Widget Lab 2', shows a data table with columns 'loc', 'temp', and 'dur'. The table contains 7 rows of data. To the right of the table are three buttons: 'Standardize Case', 'Round Decimals', and 'Standardize Units'. A 'Delete Empty Rows' button is at the bottom of the table. The game interface also includes a progress bar and a 'Data Pipeline Game' title.

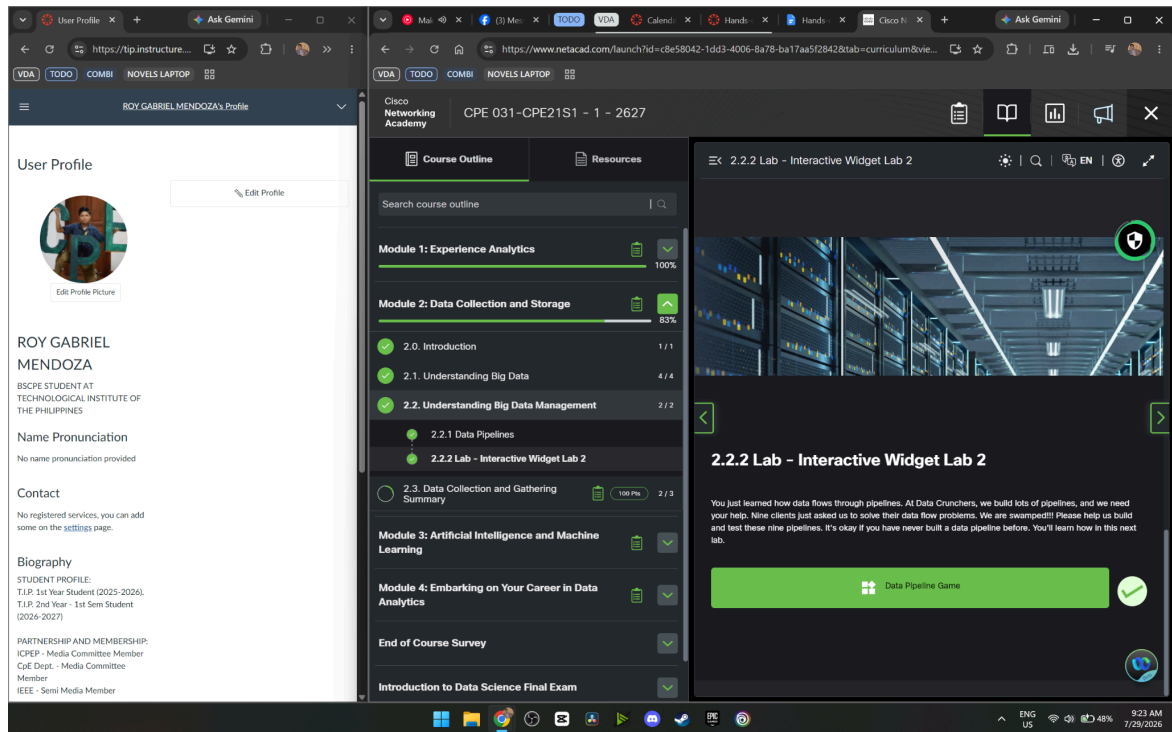
	loc	temp	dur
1	D7	98.8 F	13
2	B4	37 C	3
3	B4	38.5 C	7
4	B1	38.1 C	19
5	D4	98.8 F	20
6	B2	97 F	12
7	C0	38.9 C	10

Reflection:

In this third part of the activity, I have managed to understand that even if each errors have been fixed, There are still problems with the information that is truly the user error, for example from the picture above, The data have different Units for temperature, although It might be the decryption problem, it is very highly likely that it would be the user's input.

Overall, from this part, I have learned that even if the formatting error has been resolved, some information is still not enough for a clear understanding of it because of different **"methods"** of how the data has been put together by the user or the source.

Proof of Completion:



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: None

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: None

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: None

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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I affirm that I will not give or receive any unauthorized help on this activity/exam and that all work will be my own.